

Lecture 26

Singular Values II

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Outline and Review

5 mins

Roadmap of MATH2041

Segment 1

Solving linear systems (Lec 3-15);
Solving least squares problem (Lect 16,17)

Segment 2

[Lec 18-21]: Two relatively independent parts:
—Determinant. [Important tool!]
—Linear transformation. [Advanced math perspective of matrix]

Segment 3

Next: Lec 22-27: Eigenvalues and related.
Schedule:
—Eigenvalues. Lec 22-24
—Singular values. Lec 25-26
—Quadratic forms. Lec 27.

Today's Lecture

Outline today

- 1) **Forms** of full SVD.
- 2) Singular values; singular vectors.
- 3) **Computation** of SVD.

After today's lecture, you should be able to:

- i) Describe full SVD;
- ii) Compute singular values and singular vectors;
- iii) Tell the relation & differences of singular values/vectors and eigenvalues/eigenvectors.
- iv) Tell the relation of SVD and four fundamental subspaces.

Eigenvalues and Eigenvectors

Definition 21.1 (Eigenvalues and Eigenvectors)

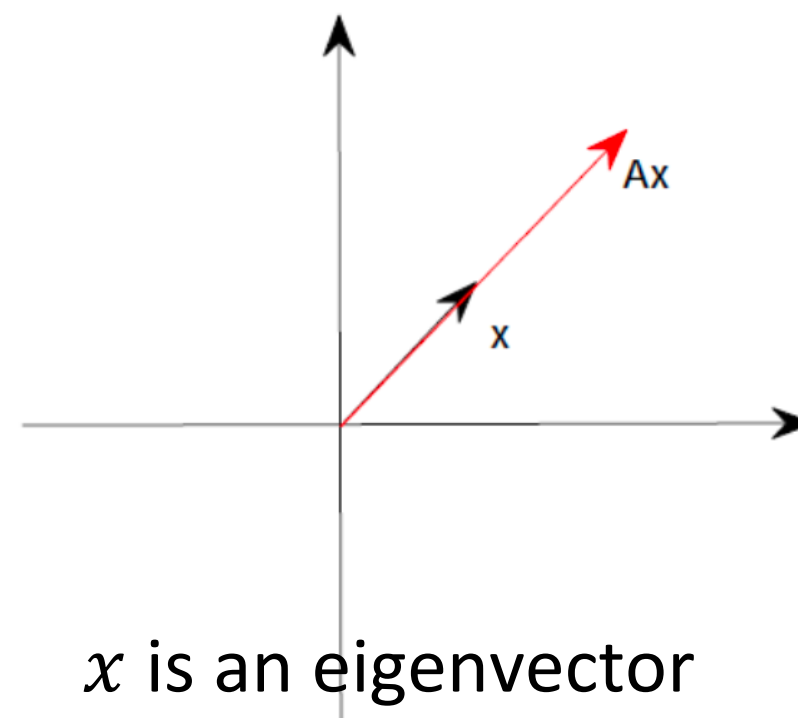
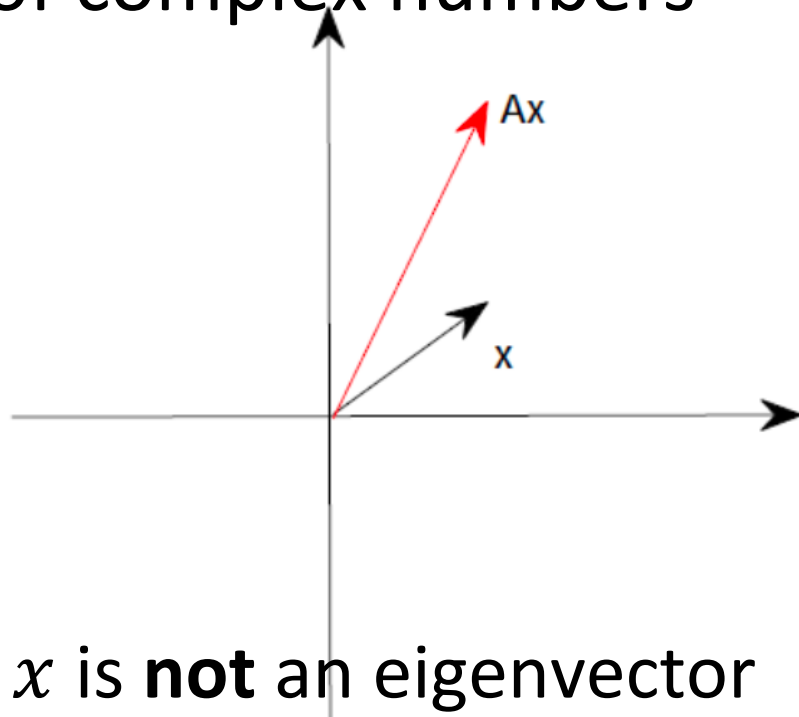
Let $A \in \mathbb{C}^{n \times n}$ be a square matrix.

If there exists a scalar λ ($\in \mathbb{R}$ or $\mathbb{C}$) and a **nonzero** vector x such that

$$Ax = \lambda x,$$

then λ is called a (real or complex) **eigenvalue** and x is called an **eigenvector** with respect to (or associated with; corresponding to) λ .

$\mathbb{C} :=$ set of complex numbers



Spectral Theorem

Definition 24.2 (Diagonalizable 可对角化)

If a square matrix A is similar to a diagonal matrix, i.e., $A = SDS^{-1}$ for some diagonal matrix D , then we say A is diagonalizable.

Theorem 23.1 [Spectral Theorem]

Any real symmetric matrix A can be written as

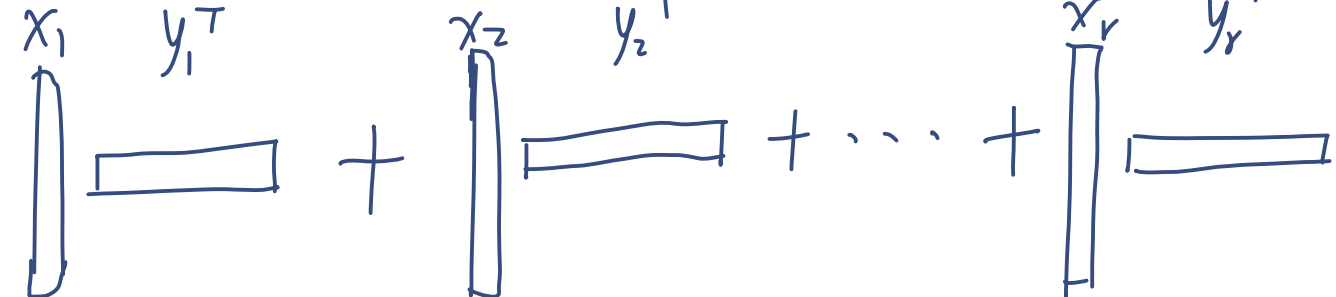
$$A = VDV^T = \sum_{j=1}^n \lambda_j \mathbf{v}_j \mathbf{v}_j^T. \quad (*)$$

where $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ is a real diagonal matrix,
 $V = [\mathbf{v}_1, \dots, \mathbf{v}_n]$ is a real orthogonal matrix.

(*): Eigenvalue decomposition (EVD) or eigendecomposition of A .

Rank-r Matrices

Proposition 25.1 Suppose $A = [\mathbf{a}_1, \dots, \mathbf{a}_n] \in \mathbb{R}^{m \times n}$ has rank r . Then

i) $\mathbf{A} = \sum_{j=1}^r \mathbf{x}_j \mathbf{y}_j^T$, 

where $\mathbf{x}_j$'s are linearly independent, $\mathbf{y}_j$'s are linearly independent.

ii) $\mathbf{A} = \mathbf{X}\mathbf{Y}^T$,

where $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_r] \in \mathbb{R}^{m \times r}$, $\mathbf{Y} = [\mathbf{y}_1, \dots, \mathbf{y}_r] \in \mathbb{R}^{n \times r}$ have rank r .

iii) $\text{Row}(\mathbf{A}) = \text{Row}(\mathbf{Y}^T) = \text{span}\{\mathbf{y}_1, \dots, \mathbf{y}_r\}$,

$\mathbf{C}(\mathbf{A}) = \mathbf{C}(\mathbf{X}) = \text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_r\}$,

$\text{Row}(\mathbf{A}) \sim \mathbf{y}_i^T$, in the expression $\sum_i x_i \mathbf{y}_i^T$, $\mathbf{y}_i^T$ is "row form".

Compact SVD

Theorem 25.1 [Compact SVD and vector form]

Any matrix $A \in \mathbb{R}^{m \times n}$ with rank r can be written as

$$A = \sum_{j=1}^r \sigma_j \mathbf{u}_j \mathbf{v}_j^T = U_r \Sigma_r V_r^T, \quad (*)$$

Where $U_r = [\mathbf{u}_1, \dots, \mathbf{u}_r] \in \mathbb{R}^{m \times r}$ has orthogonal unit-norm columns,
 $V_r = [\mathbf{v}_1, \dots, \mathbf{v}_r] \in \mathbb{R}^{n \times r}$ has orthogonal unit-norm columns,
 $\Sigma_r = \text{diag}(\sigma_1, \dots, \sigma_r) \in \mathbb{R}_+^{r \times r}$ is a diagonal matrix.

Here $\mathbb{R}_+$ is the set of non-negative real numbers.

Definition (Frobenius norm)

For a matrix $A = (a_{ij})_{m \times n}$, $\|A\|_F \triangleq \sqrt{\sum_{i=1}^m \sum_{j=1}^n a_{i,j}^2}$

Claim: $\|A\|_F = \sqrt{\sigma_1^2 + \dots + \sigma_r^2}$.

Approximate Compression of Data Matrix

In practice, the rank r is often large, but can
throw away small components as approximation!

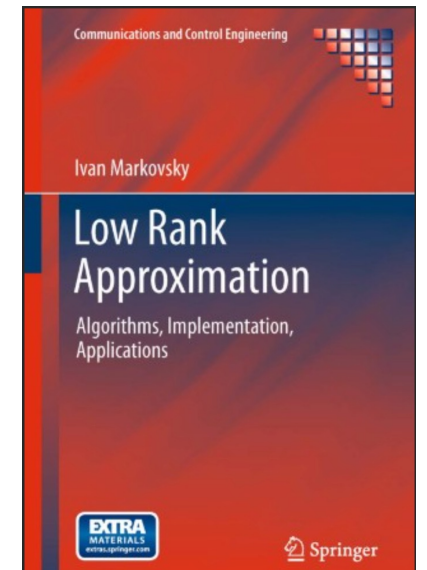
Example: Compact SVD

$$A = \mathbf{u}_1 \mathbf{v}_1^T + 0.01 \mathbf{u}_2 \mathbf{v}_2^T ,$$

Then $\hat{A} = \mathbf{u}_1 \mathbf{v}_1^T$ is a “good approximation” of A .

i.e., “approximation error”

$$\|A - \hat{A}\|_F = 0.01.$$



Part I SVD from Linear Transformation Perspective

35 mins

Part I.1 Mapping Basis of Row Space to Column Space

15 mins

Preparation: Size of Matrix

$A: R^s \rightarrow R^t$, what is the size of A ?

Method 1 A maps $x \in R^s$ to $Ax = y \in R^t$.

$\uparrow$
 $R^{t \times s}$

$\downarrow \quad \downarrow \quad \downarrow$
 $t \times s \quad s \times 1 \quad t \times 1$
 $\swarrow$
match
 $t \times s$

Method 2 (easier to memorize)

input space output space
 $A: R^s \rightarrow R^t$

$A \in R^{t \times s}$

$s = \dim$ of input space = d_{in}
 $t = \dim$ of output space = d_{out}

$\begin{pmatrix} Ax = y \\ x \cdot A \end{pmatrix}$

Thus $A \in R^{d_{out} \times d_{in}}$. (memorize: $d_{out} \times d_{in}$)

Examples

If A is $R^{2 \times 3}$,
then $A: R^3 \rightarrow R^2$.

If A is $R^{3 \times 2}$,
then $A: R^2 \rightarrow R^3$.

If A is $R^{3 \times 5}$,
then $A: R^5 \rightarrow R^3$.

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$$A: 4 \times 10.$$

$$\mathbb{R}^{10} \rightarrow \mathbb{R}^4$$

Compact SVD & Linear Transformation

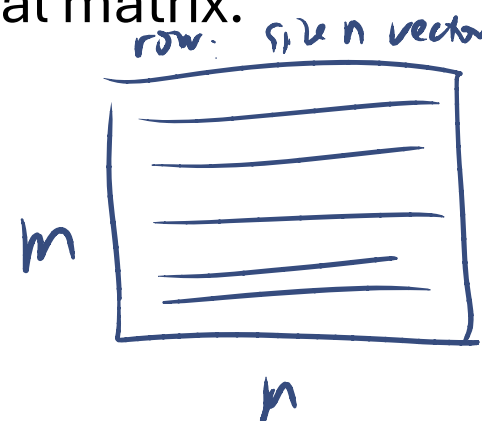
Any matrix $A \in \mathbb{R}^{m \times n}$ with rank r can be written as

$$A = \sum_{j=1}^r \sigma_j \mathbf{u}_j \mathbf{v}_j^T = U_r \Sigma_r V_r^T, \quad (*)$$

Where $U_r = [\mathbf{u}_1, \dots, \mathbf{u}_r] \in \mathbb{R}^{m \times r}$ has orthogonal unit-norm columns,
 $V_r = [\mathbf{v}_1, \dots, \mathbf{v}_r] \in \mathbb{R}^{n \times r}$ has orthogonal unit-norm columns,
 $\Sigma_r = \text{diag}(\sigma_1, \dots, \sigma_r) \in \mathbb{R}_+^{r \times r}$ is a diagonal matrix.

$$\text{Row}(A) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_r) \subseteq \mathbb{R}^n.$$

$$\text{C}(A) = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_r) \subseteq \mathbb{R}^m.$$



Recall: $f(x) = Ax: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation.

Natural thought: Let's look at $A \mathbf{v}_i$.

$$\mathbf{u}_i^T \mathbf{u}_j = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}, \quad \mathbf{v}_i^T \mathbf{v}_j = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

$$A \mathbf{v}_i = \sum_{j=1}^r \sigma_j \mathbf{u}_j \mathbf{v}_j^T \mathbf{v}_i = \sigma_i \mathbf{u}_i (\mathbf{v}_i^T \mathbf{v}_i) = \sigma_i \mathbf{u}_i$$

$\mathbf{v}_j^T \mathbf{v}_i = 0, \forall j \neq i$

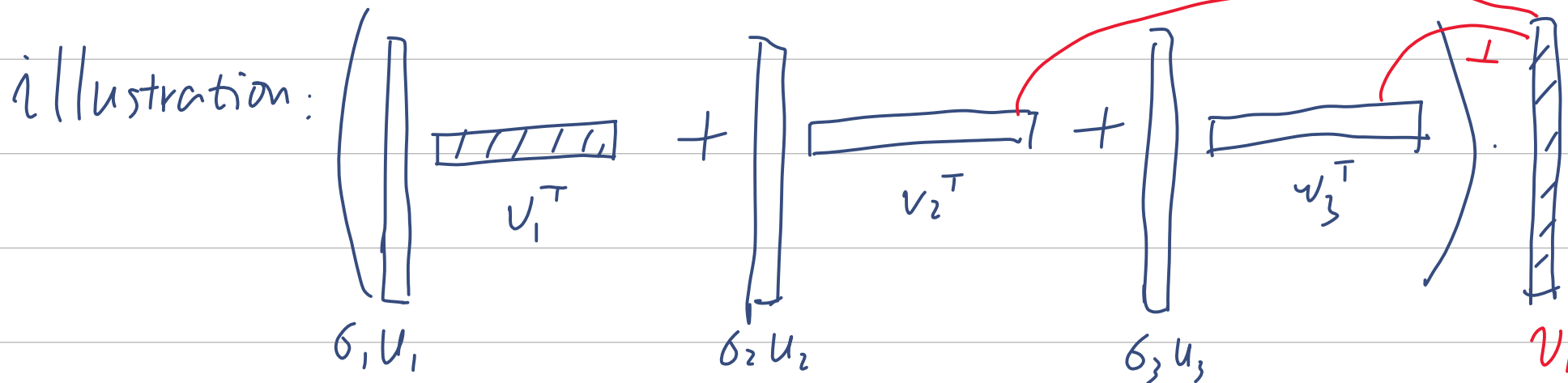
~~$\sigma_j \mathbf{u}_j$~~

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$$A v_i = \sum_{j=1}^r \sigma_j u_j v_j^T v_i$$

e.g. $r=3, i=1$.

$$\begin{aligned} A v_i &= (\sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \sigma_3 u_3 v_3^T) v_i \\ &= \sigma_1 u_1 (v_1^T v_i) + \sigma_2 u_2 (\underbrace{v_2^T v_i}_{\substack{1 \\ 0}}) + \sigma_3 u_3 (\underbrace{v_3^T v_i}_{\substack{1 \\ 0}}) \\ &= \sigma_1 u_1 \end{aligned}$$



Mapping Row Space to Column Space

$$A = U_r \Sigma_r V_r^T = \sum_{i=1}^r \sigma_i u_i v_i^T.$$

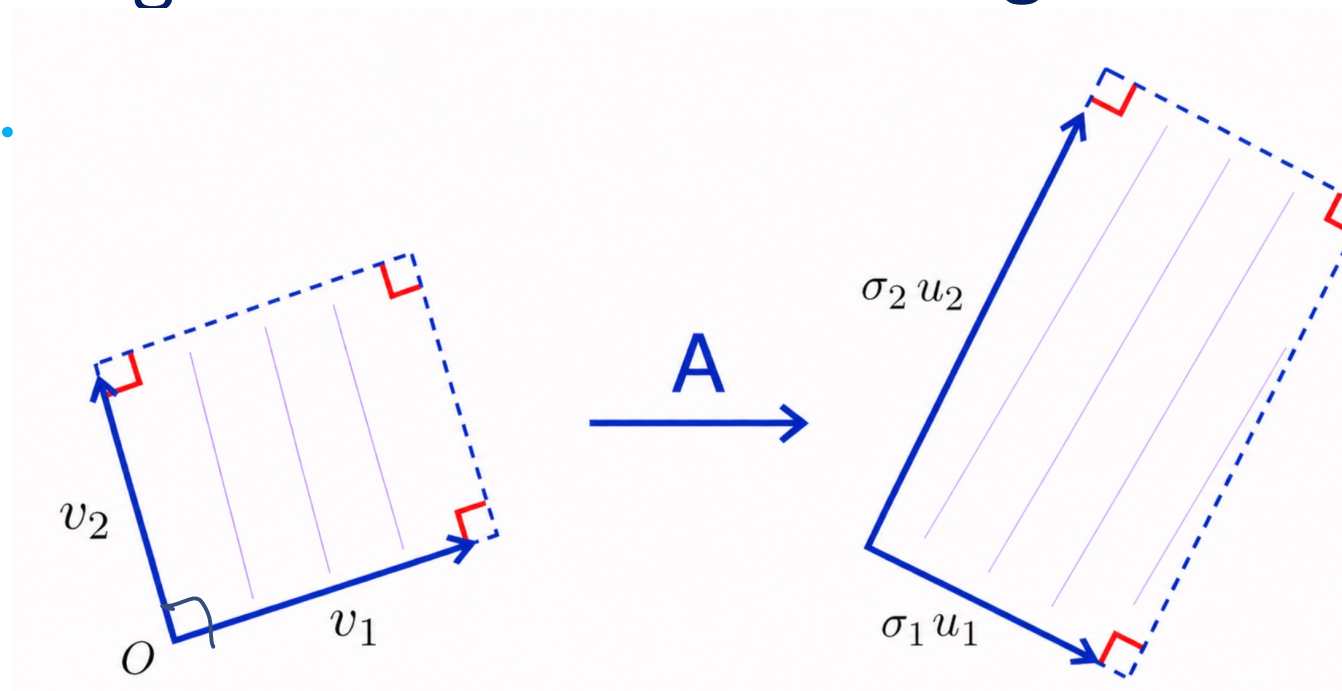
$$Av_1 = \sigma_1 u_1, \quad Av_2 = \sigma_2 u_2, \quad \dots, \quad Av_r = \sigma_r u_r.$$

input direction v_i
orthogonal

$\xrightarrow{A}$

output direction u_i
orthogonal ; with stretching factor σ_i .

E.g. $r = 2$.



Matrix Form

Recall

$$\underline{U}_r = [\mathbf{u}_1, \dots, \mathbf{u}_r] \in \mathbb{R}^{m \times r}$$

$$\underline{V}_r = [\mathbf{v}_1, \dots, \mathbf{v}_r] \in \mathbb{R}^{n \times r}$$

$\underline{\Sigma}_r = \text{diag}(\sigma_1, \dots, \sigma_r) \in \mathbb{R}_+^{r \times r}$ is a diagonal matrix.

Previous page:

$$Av_1 = \sigma_1 u_1,$$

$$Av_2 = \sigma_2 u_2,$$

$\dots,$

$$Av_r = \sigma_r u_r.$$

matrix form

$$A \underline{V}_r = \underline{U}_r \underline{\Sigma}_r$$

$$Av_i = \lambda_i v_i, i=1, \dots, n.$$

$$\begin{array}{c} \Updownarrow \\ A \cdot V = V \cdot D \end{array}$$

Part I.2 Mapping Whole Space and Deriving Full SVD: Case of 4×3 Matrix

15 mins

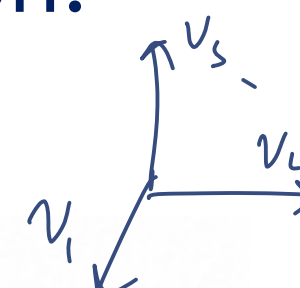
Mapping Whole Space: 3D Case

3D case: Consider 4×3 matrix A , with rank 2.
 $f(x) = Ax: \mathbb{R}^3 \rightarrow \mathbb{R}^4$ is a linear transformation.

$$Av_1 = \sigma_1 u_1, \quad Av_2 = \sigma_2 u_2$$

The vectors v_1, v_2 form an orthonormal basis of $\text{Row}(A)$.

To describe all of $\mathbb{R}^3$, choose one more unit vector $v_3 \perp \text{Row}(A)$.



Then what happens to v_3 ?

Now v_1, v_2, v_3 form an orthonormal basis of $\mathbb{R}^3$.

Since $v_3 \perp \text{Row}(A)$, we have $v_3 \in N(A)$ [some space related to A]

Hence $Av_3 = 0$

Input space			Output space
$\mathbb{R}^3$	v_1	$\xrightarrow{A}$	$\sigma_1 u_1$
	v_2	$\xrightarrow{A}$	$\sigma_2 u_2$
	v_3	$\xrightarrow{A}$	0

Four Fundamental Subspaces v.s. full SVD

Consider the big picture:

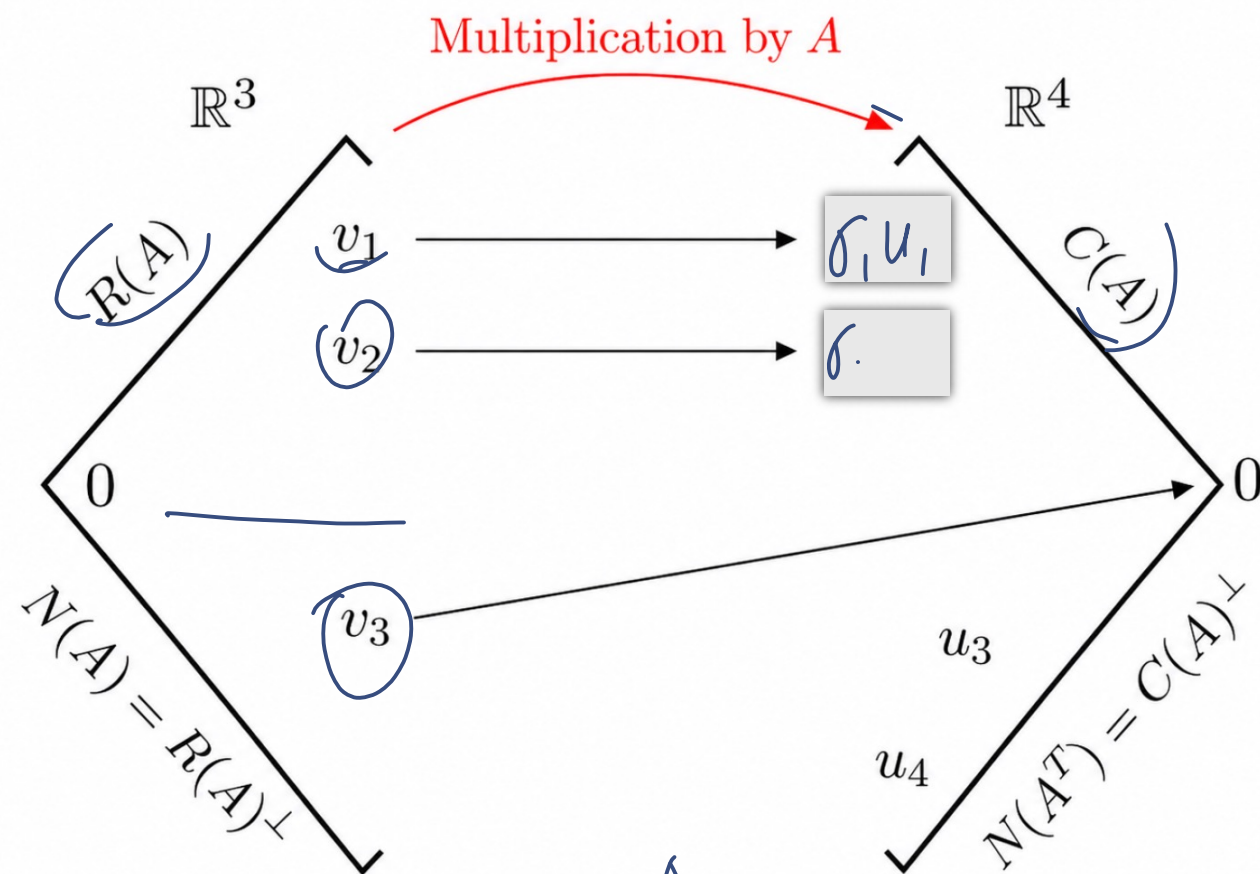
Add $\mathbf{u}_3, \mathbf{u}_4$, s.t. $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4$ form an orthonormal basis of $\mathbb{R}^4$.

Input space

$$\mathbb{R}^3 = \text{Row}(A) \cup N(A)$$

Output space

$$\mathbb{R}^4 = C(A) \cup N(A^T)$$



New: row space $\xrightarrow{A}$ column space

Matrix Form

$$Av_1 = \sigma_1 u_1, Av_2 = \sigma_2 u_2$$

$$Av_3 = \mathbf{0}.$$

Let $V = [v_1, v_2, v_3] \in \mathbb{R}^{3 \times 3}$; an orthogonal matrix

Let $U = [u_1, u_2, u_3, u_4] \in \mathbb{R}^{4 \times 4}$; an orthogonal matrix.

$$\begin{aligned} AV_r &= U_r \Sigma_r \\ \Sigma_r &= \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix} \\ U_r &= [u_1, u_2] \\ V_r &= [v_1, v_2] \end{aligned}$$

Matrix Form:

$$AV = U\Sigma,$$

$$4 \times 3 \quad 3 \times 3 \quad 4 \times 4 \quad 4 \times 3$$

where $\Sigma =$

$$\begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Matrix Decomposition Form and Vector Form

Compact SVD: $A = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T = U_2 \Sigma_2 V_2^T$.

Let $V = [\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3] \in \mathbb{R}^{3 \times 3}$; an orthogonal matrix

Let $U = [\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4] \in \mathbb{R}^{4 \times 4}$; an orthogonal matrix.

Matrix Form:

$$A V = U \Sigma \Rightarrow A = U \Sigma V^T.$$

$4 \times 3 \quad 4 \times 4 \quad 4 \times 3 \quad 3 \times 3$

Vector Form:

$$A = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T + \sigma_3 \mathbf{u}_3 \mathbf{v}_3^T, \quad \Sigma = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \\ 0 & 0 & 0 \end{bmatrix}$$

where $\sigma_3 = 0$.

Part 1.3 Mapping Whole Space and Deriving Full SVD: $m \times n$ Matrix

5 mins

Mapping Whole Space: General Case

$f(x) = Ax: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation.

Its input space is the whole $\mathbb{R}^n$,

not just $v_1, \dots, v_r$, basis of $\text{Row}(A)$.

Add $v_{r+1}, \dots, v_n$, s.t. $v_1, \dots, v_n$ form an orthonormal basis of $\mathbb{R}^n$.

Then $Av_1 = \sigma_1 u_1, \dots, Av_r = \sigma_r u_r$;

$$Av_{r+1} = \dots = Av_n = \mathbf{0}.$$

Four Fundamental Subspaces v.s. full SVD

Consider the big picture:

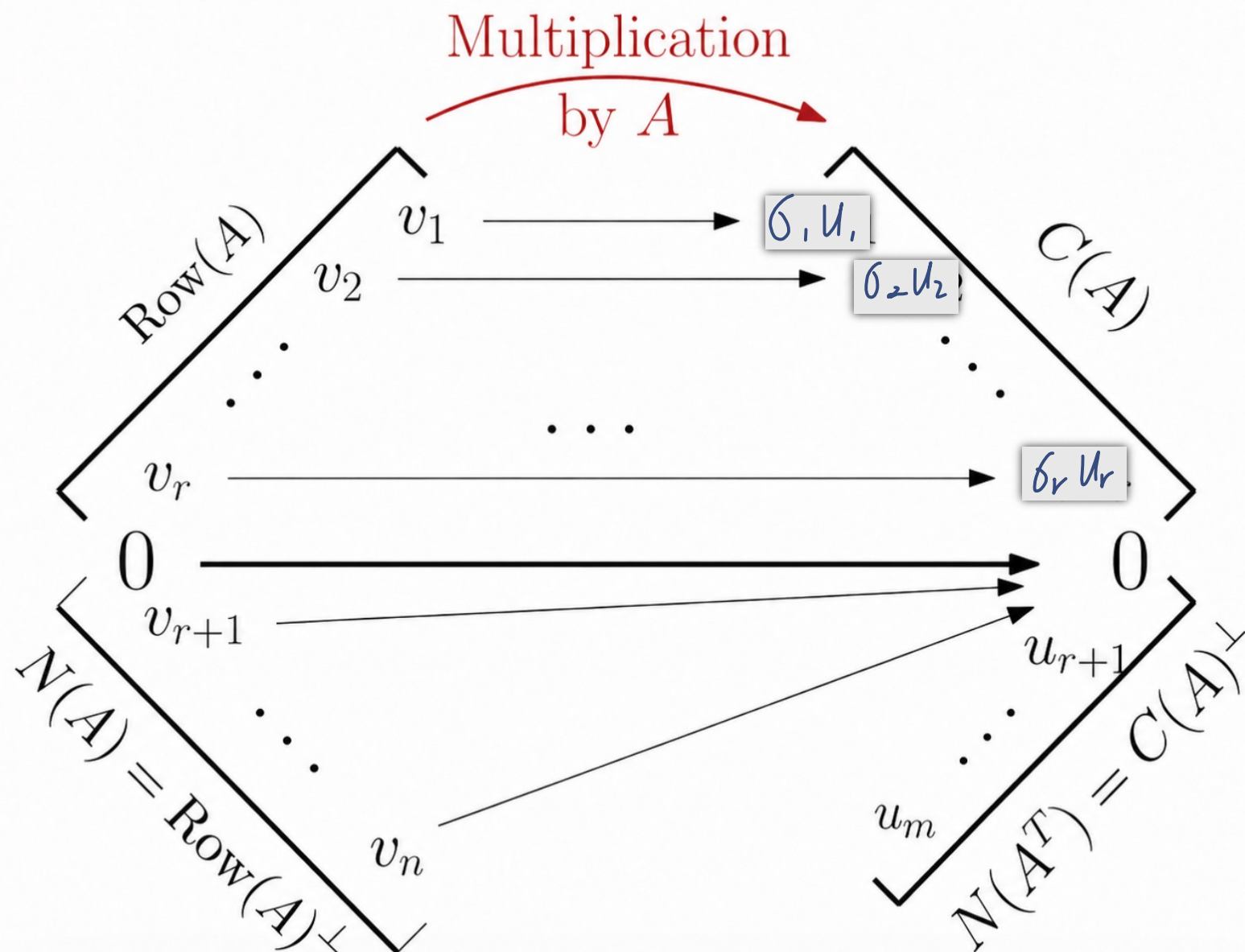
Add $\mathbf{u}_{r+1}, \dots, \mathbf{u}_m$, s.t. $\mathbf{u}_1, \dots, \mathbf{u}_m$ form an orthonormal basis of $\mathbb{R}^m$.

Input space

$$\mathbb{R}^n = \text{Row}(A) \cup N(A)$$

Output space

$$\mathbb{R}^m = C(A) \cup N(A^T)$$



Matrix Form

$$A\mathbf{v}_1 = \sigma_1 \mathbf{u}_1, \dots, A\mathbf{v}_r = \sigma_r \mathbf{u}_r ;$$

$$A\mathbf{v}_{r+1} = \dots = A\mathbf{v}_n = \mathbf{0}.$$

Let $V = [\mathbf{v}_1, \dots, \mathbf{v}_n] \in \mathbb{R}^{n \times n}$; an orthogonal matrix

Let $U = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m] \in \mathbb{R}^{m \times m}$; an orthogonal matrix.

Let $\sigma_{r+1} = \dots = \sigma_{\min(n,m)} = 0$

$$\text{Matrix Form: } \mathbf{A} \mathbf{V} = \mathbf{U} \Sigma, \quad \Rightarrow A = U \Sigma V^T.$$

$$m \times n \quad n \times n \quad m \times m \quad m \times n$$

where

$$\Sigma =$$

$$\begin{bmatrix} \sigma_1 & & & & & & 0 \\ & \ddots & & & & & \\ & & \sigma_r & & & & \\ & & & \sigma_{r+1} & & & \\ & & & & \ddots & & \\ & & & & & \sigma_n & \\ 0 & & & & & & \vdots \\ \vdots & & & & & & \\ 0 & & & & & & \vdots \end{bmatrix}$$

$$m \geq n.$$

or

$$\begin{bmatrix} \sigma_1 & & & 0 & & \\ & \ddots & & & & \\ & & \sigma_m & & & \\ 0 & & & & & \\ & & & & & \vdots \end{bmatrix}$$

$$m < n$$

Four Fundamental Subspaces

Let $A = [u_1, \dots, u_r, u_{r+1}, \dots, u_m]$

$$\left[\begin{array}{ccc|c} \sigma_1 & & (0) & 0 \\ & \ddots & & \\ (0) & & \sigma_r & 0 \\ \hline & 0 & & 0 \end{array} \right] \begin{bmatrix} v_1^T \\ \vdots \\ v_r^T \\ v_{r+1}^T \\ \vdots \\ v_n^T \end{bmatrix}.$$

Column space $C(A)$: dimension r , orthonormal basis $\{u_1, u_2, \dots, u_r\}$

Null space: dimension $n - r$, basis $\{v_{r+1}, \dots, v_n\}$

Row space: dimension r , basis $\{v_1, \dots, v_r\}$

Left - null space: dimension $m - r$, basis $\{u_{r+1}, \dots, u_m\}$

Part II Full SVD

5 mins

Rectangular-Diagonal Matrix

For simplicity of discussion, we introduce a term below.

Rectangular diagonal matrix $\Sigma \in \mathbb{R}^{m \times n}$ is a matrix of the form

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_n \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{bmatrix} \quad \text{or} \quad \Sigma = \begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_m \end{bmatrix} \begin{matrix} \left[\begin{matrix} 0 & \dots & 0 \\ 0 & \dots & 0 \\ 0 & \dots & 0 \end{matrix} \right] \\ \left[\begin{matrix} 0 & \dots & 0 \\ 0 & \dots & 0 \end{matrix} \right] \end{matrix}$$

Denote as $\Sigma = \text{rec-diag}(\sigma_1, \dots, \sigma_{\min(m,n)})_{m \times n}$

This is not that widely used;
It is convenient for SVD.

E.g. $\begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix} = \text{rec-diag}(4, 2)_{3 \times 2}$

$$\begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \text{rec-diag}(4, 2, 0)_{3 \times 4}$$

$$\begin{bmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \text{rec-diag}(4, 2, 0)_{3 \times 3}$$

Think: what 0 in the matrix
shall be included in rec-diag (...),
and what 0 shall not?

SVD Theorem

Theorem 26.1 (full SVD)

Any matrix $A \in \mathbb{R}^{m \times n}$ can be written as

$$A = U \Sigma V^T = \sum_{i=1}^{\min(m,n)} \sigma_i \mathbf{u}_i \mathbf{v}_i^T \quad (*)$$

where $U \in \mathbb{R}^{m \times m}$, $V \in \mathbb{R}^{n \times n}$ are two **orthogonal matrices**,
 $\Sigma = \text{rec-diag}(\sigma_1, \dots, \sigma_{\min(n,m)}) \in \mathbb{R}_+^{m \times n}$ is a **rectangular diagonal matrix**.

Definition (singular values 奇异值; singular vectors 奇异向量):

The columns of $U \in \mathbb{R}^{m \times m}$ are called the **left singular vectors** of A ;

The columns of $V \in \mathbb{R}^{n \times n}$ called the **right singular vectors** of A ;

$\sigma_1, \dots, \sigma_{\min(m,n)}$ are called the **singular values** of A .

Remark: SVD is short for “singular value decomposition”.

SVD Examples

Example 1:

Matrix:

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

SVD Components:

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad V = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ \frac{2}{\sqrt{6}} & 0 & -\frac{1}{\sqrt{3}} \end{bmatrix}$$

Verification:

$$U\Sigma V^T = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} = A$$

SVD Examples

Example 2:

Matrix:

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

SVD Components:

$$U = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Verification:

$$U\Sigma V^T = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} = A$$

Full SVD v.s. Compact SVD

$$A = U\Sigma V^T \quad \text{v.s.} \quad A = \underline{U}_r \underline{\Sigma}_r \underline{V}_r^T$$

Compared to **compact SVD**:

Add $\mathbf{u}_{r+1}, \dots, \mathbf{u}_m$, s.t. $U = [U_r, \mathbf{u}_{r+1}, \dots, \mathbf{u}_m] \in \mathbb{R}^{m \times m}$ is an orthogonal matrix;

Add $\mathbf{v}_{r+1}, \dots, \mathbf{v}_n$, s.t. $V = [V_r, \mathbf{v}_{r+1}, \dots, \mathbf{v}_n] \in \mathbb{R}^{n \times n}$ is an orthogonal matrix.

$\mathbf{u}_1, \dots, \mathbf{u}_m$ form an orthonormal basis of $\mathbb{R}^m$;

$\mathbf{v}_1, \dots, \mathbf{v}_n$ form an orthonormal basis of $\mathbb{R}^n$.

Add $\sigma_{r+1} = \dots = \sigma_{\min(n,m)} = 0$.

Compact SVD & full SVD are equivalent when

$r = m = n$, i.e. square full rank matrix.

Full SVD v.s. Compact SVD

Full SVD: Any matrix $A \in \mathbb{R}^{4 \times 3}$ can be written as

$$A = U \Sigma V^T,$$

4×3 4×4 4×3 3×3

Orthogonal matrix Rec-Diagonal Orthogonal matrix

Compact SVD: Any matrix $A \in \mathbb{R}^{4 \times 3}$ with rank 2 can be written as

$$A = U_r \Sigma_r V_r^T,$$

4×3 4×2 2×2 2×3

Columns Orthogonal, Unit-norm Diagonal Orthogonal matrix

Part III Singular Vectors and Singular Values

20 mins

Part III.1 Mapping $A^T x$

6 mins

Why Consider $\mathbf{u}_{r+1}, \dots, \mathbf{u}_m$?

Suppose the full SVD is:

$$A = U \Sigma V^T = \sum_{i=1}^{\min(m,n)} \sigma_i \mathbf{u}_i \mathbf{v}_i^T$$

$$A\mathbf{v}_1 = \sigma_1 \mathbf{u}_1, \dots, A\mathbf{v}_r = \sigma_r \mathbf{u}_r ;$$
$$A\mathbf{v}_{r+1} = \dots = A\mathbf{v}_n = \mathbf{0}.$$

One may wonder:

Why do we even need to consider $\mathbf{u}_{r+1}, \dots, \mathbf{u}_m$?

Just for getting a (square) orthogonal matrix U ?

It seems $\mathbf{u}_{r+1}, \dots, \mathbf{u}_m$ do NOT have physical meaning.
Well...they do have...

View of Transpose

Suppose the full SVD is:

$$A = U \Sigma V^T = \sum_{i=1}^{\min(m,n)} \sigma_i \mathbf{u}_i \mathbf{v}_i^T$$

Take transpose:

$$A^T = \underbrace{V}_{\text{orthogonal matrix}} \underbrace{\Sigma^T}_{\text{rec-diagonal}} \underbrace{U^T}_{\text{orthogonal matrix}} = \sum_{i=1}^{\min(m,n)} \sigma_i \boxed{v_i u_i^T}$$

Similarly, we have:

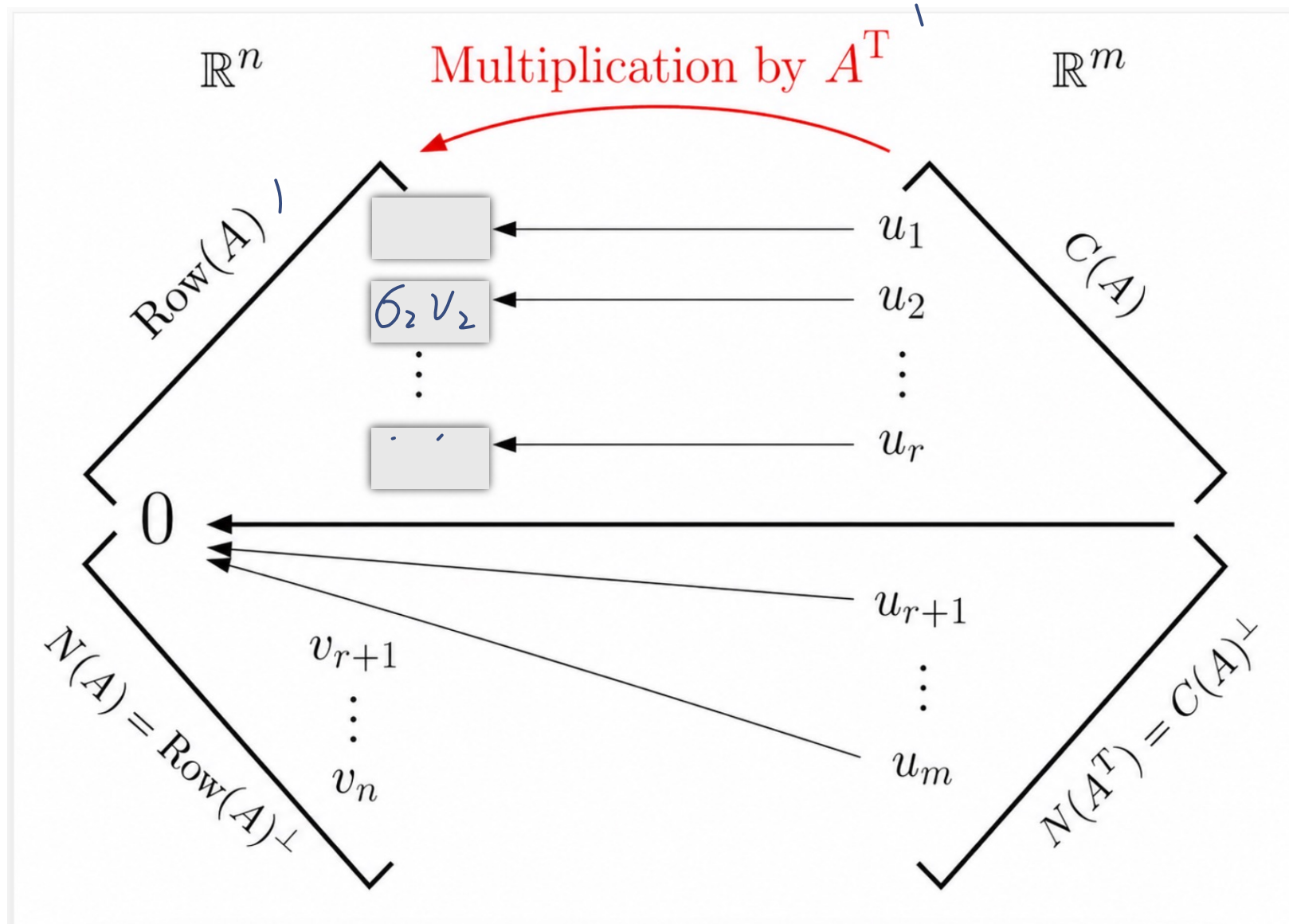
$$\begin{aligned} A^T \boxed{u_1} &= \boxed{\sigma_1 v_1}, \dots, A^T \boxed{u_r} = \boxed{\sigma_r v_r}; \\ A^T \boxed{u_{r+1}} &= \dots = A^T \boxed{u_m} = \mathbf{0}. \end{aligned}$$

Mapping $A^T x$

$$A = \begin{bmatrix} \vdots \\ u \end{bmatrix} \begin{bmatrix} v^T \end{bmatrix}$$

We can draw a similar figure:

$$A^T = \begin{bmatrix} v \\ \vdots \end{bmatrix} \begin{bmatrix} u^T \end{bmatrix}$$



A^T maps column space to row space; maps $N(A^T)$ to 0.

Part III.2

Relation to $A^T A$ and AA^T

14 mins

Combining Two Equations?

We have: $A\mathbf{v}_1 = \sigma_1 \mathbf{u}_1, \dots, A\mathbf{v}_r = \sigma_r \mathbf{u}_r$

We also have: $A^T \mathbf{u}_1 = \sigma_1 \mathbf{v}_1, \dots, A^T \mathbf{u}_r = \sigma_r \mathbf{v}_r$

Combine them?

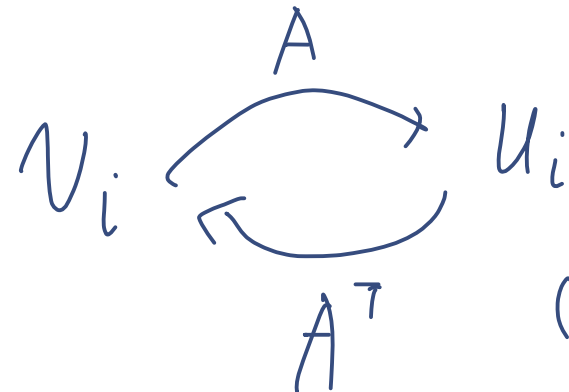
multiply
 A^T

$$A\mathbf{v}_i = \sigma_i \mathbf{u}_i$$

$$A^T \mathbf{u}_i = \sigma_i \mathbf{v}_i$$

$\Rightarrow$

$$\begin{aligned} \underline{A^T A} \mathbf{v}_i &= A^T (\sigma_i \mathbf{u}_i) \\ &= \sigma_i (A^T \mathbf{u}_i) \\ &= \sigma_i (\sigma_i \mathbf{v}_i) \\ &= \sigma_i^2 \mathbf{v}_i \end{aligned}$$



(ignore scalar
coefficients)

Eigenvalues and Eigenvectors

Observation 1:

$$A^T A \mathbf{v}_i = \sigma_i^2 \mathbf{v}_i, i = 1, \dots, r.$$

This means:

$\sigma_1^2, \dots, \sigma_r^2$ are *eigenvalues* of $A^T A$;
 $\mathbf{v}_1, \dots, \mathbf{v}_r$ are the corresponding *eigenvectors* of $A^T A$.

Suppose eigenvalues of $A^T A$ are:

$$\lambda_1 \geq \dots \geq \lambda_r > \lambda_{r+1} = \dots = \lambda_n = 0.$$

Then

$$\sigma_1 = \sqrt{\lambda_1}, \dots, \sigma_r = \sqrt{\lambda_r} \text{ are singular values of } A. \quad (3.1)$$

$\sigma_{r+1} = \dots = \sigma_{\min\{m,n\}} = 0$, so can write a similar form:

$$\sigma_{r+1} = \lambda_{r+1}, \dots, \sigma_{\min\{m,n\}} = \sqrt{\lambda_{\min\{m,n\}}} \text{ are singular values of } A. \quad (3.2)$$

Combining Two Equations: Another Way

Combine them?

$$\left. \begin{array}{l} \underline{A} \underline{v}_i = \underline{\sigma}_i \underline{u}_i \\ A(A^T \underline{u}_i) = A(\sigma_i \underline{v}_i) \end{array} \right\} \Rightarrow \begin{array}{l} A(A^T \underline{u}_i) = A(\sigma_i \underline{v}_i) \\ = \sigma_i (A \underline{v}_i) \\ = \sigma_i (\sigma_i \underline{u}_i) \\ = \sigma_i^2 \underline{u}_i \end{array}$$

Observation 2: $AA^T \underline{u}_i = \sigma_i^2 \underline{u}_i, i = 1, \dots, r.$

This means:

$\sigma_1^2, \dots, \sigma_r^2$ are *eigenvalues* of AA^T (3.3)

$\underline{u}_1, \dots, \underline{u}_r$ are the corresponding *eigenvectors* of AA^T (3.4)

(3.1) & (3.3) $\rightarrow$

$\sigma_1^2, \dots, \sigma_r^2$ are simultaneously nonzero *eigenvalues* of AA^T and $A^T A$.

Singular values

For $A \in \mathbb{R}^{m \times n}$, $A = U\Sigma V^T$ and $\text{rank}(A)=r$, one has

$\lambda_1 \geq \cdots \geq \lambda_r > \lambda_{r+1} = \cdots = \lambda_n = 0$ are the eigenvalues of $\overbrace{A^T A}^{n \times n}$.

$\lambda_1 \geq \cdots \geq \lambda_r > \lambda_{r+1} = \cdots = \lambda_m = 0$ are the eigenvalues of $\underbrace{AA^T}_{m \times m}$.

(1) If $m \geq n$, as shown in above, let $\sigma_i = \sqrt{\lambda_i}$ ($i = 1, \dots, n$), then $\sigma_1 \geq \cdots \geq \sigma_r > \sigma_{r+1} = \cdots = \sigma_n = 0$ (with $n - r$ zeros) are the **singular values**.

If $m < n$, let $\sigma_i = \sqrt{\lambda_i}$ ($i = 1, \dots, m$), then $\sigma_1 \geq \cdots \geq \sigma_r > \sigma_{r+1} = \cdots = \sigma_m = 0$ (with $m - r$ zeros) are the **singular values**.

The number of singular values is $\min(m, n)$.

Singular Vectors in full SVD

Three parts that form U,V:

Part 1 (pairing):

↙ Eigenvectors for non-zero eigenvalues of $A^T A$
 $v_1, \dots, v_r$

They lead to $u_1 = \frac{Av_1}{\sigma_1}, \dots, u_r = \frac{Av_r}{\sigma_r}$.

Part 2 (null space):

Orthonormal basis of $N(A)$: $v_{r+1}, \dots, v_n$

Part 3 (left-null space):

Orthonormal basis of $N(A^T)$: $u_{r+1}, \dots, u_m$.

Properties of Singular Values & Singular Vectors

Property 26.1 (relation with eigenvalues & eigenvectors)

Left singular vectors are **eigenvectors of $AA^T \in \mathbb{R}^{m \times m}$** .

Right singular vectors are **eigenvectors ^{of} $A^T A \in \mathbb{R}^{n \times n}$** .

$\sigma_1, \dots, \sigma_{\min(m,n)}$, Singular values of A are **square roots of eigenvalues** of both AA^T and $A^T A$.

Property 26.2 [singular-equation-pairs]

Suppose $A \in \mathbb{R}^{m \times n}$, the SVD is $A = U \Sigma V^T$, where $\Sigma = \text{rec-diag}(\sigma_1, \dots, \sigma_{\min(m,n)})$, $V = [v_1, \dots, v_n]$, $U = [u_1, \dots, u_m]$, in which $v_j \in \mathbb{R}^{n \times 1}$, $u_j \in \mathbb{R}^{m \times 1}$, $\sigma_j \in \mathbb{R}$.

Let $k = \min\{m, n\}$. Then

$$Av_1 = \sigma_1 u_1, \dots, Av_k = \sigma_k u_k$$

$$A^T u_1 = \sigma_1 v_1, \dots, A^T u_k = \sigma_k v_k.$$

Part IV

Computing SVD

5 mins

Compute full SVD

Problem: Find the full SVD of the matrix A .

Step 1: Find eigenvalues of $A^T A$.

Suppose they are: $\lambda_1, \dots, \lambda_r > 0$; and $\lambda_{r+1} = \dots = \lambda_n = 0$.

Denote $\sigma_j = \sqrt{\lambda_j}$, where $j = 1, \dots, \min(m, n)$.

Step 2: Find right singular vectors.

Step 2.1: Find r unit eigenvectors of $A^T A$ corresponding to $\lambda_1, \dots, \lambda_r$.

--If eigenvalues are distinct, then they are automatically orthogonal;

--If there are repeated eigenvalues, find orthonormal basis of the eigenspace.

Step 2.2: Find an orthonormal basis of $N(A)$, denote as $v_{r+1}, \dots, v_n$.

Step 3: Find left singular vectors of A .

Step 3.1: Define Left singular vectors $u_i = \frac{Av_i}{\sigma_i}$ for $i = 1, \dots, r$.

Step 3.2: Find an orthonormal basis of $N(A^T)$, denote as $u_{r+1}, \dots, u_m$.

Write $\Sigma = \text{rec-diag}(\sigma_1, \dots, \sigma_{\min(m,n)})$, $U = (u_1, \dots, u_m)$, $V = (v_1, \dots, v_n)$.

Conclusion: $A = U\Sigma V^T$ where U, Σ, V are defined above.

Example: Compute SVD

Problem 1: Find the SVD of the matrix $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}$.

Solution:

Step 1: Find **eigenvalues of $A^T A$** (which is 2 by 2, where $2 = \min(m,n)$)

$$A^T A = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

Eigenvalues of $A^T A$ are 4 and 0. Then $\sigma_1 = 2$, $\sigma_2 = 0$.

Example: Compute SVD

Problem 1: Find the SVD of the matrix $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}$.

Solution:

Step 1: Find **eigenvalues of $A^T A$** (which is 2 by 2, where $2 = \min(m,n)$)

$$A^T A = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

Eigenvalues of $A^T A$ are 4 and 0. Then $\sigma_1 = 2$, $\sigma_2 = 0$.

Step 2: Find (unit) **eigenvectors of $A^T A$** , i.e. right singular vectors of A.

A unit eigenvector corresponding to $\lambda_1 = 4$ is $v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

A unit eigenvector corresponding to $\lambda_2 = 0$ is $v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Step 3: Define $u_1 = \frac{Av_1}{\sigma_1}$, $u_2 = \frac{Av_2}{\sigma_2} = \frac{Av_2}{0}$

Wrong method!

Problem 1: Step 3 and Remark

Step 3: Find left singular vectors of A.

Now $\sigma_1 = \sqrt{\lambda_1} = 2$ so

$$\mathbf{u}_1 = \frac{1}{\sigma_1} A \mathbf{v}_1 = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

Left singular vectors $\mathbf{u}_i = \frac{A \mathbf{v}_i}{\sigma_i}$

An orthonormal basis for $N(A^T)$ is

$$\{\mathbf{u}_2, \mathbf{u}_3\} = \left\{ \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Rest: orthonormal basis
of left null space.

Conclusion: The full SVD of A is

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix} = A = U \Sigma V^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

Problem 1: Verification

Remark: Verify your SVD by checking matrix product

$$U \Sigma V^T = A$$

$$\begin{aligned} U \Sigma V^T &= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \\ &= \begin{bmatrix} \sqrt{2} & 0 \\ \sqrt{2} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \\ &= \begin{bmatrix} \sqrt{2} \cdot \frac{1}{\sqrt{2}} + 0 \cdot \frac{1}{\sqrt{2}} & \sqrt{2} \cdot \frac{1}{\sqrt{2}} + 0 \cdot \left(-\frac{1}{\sqrt{2}}\right) \\ \sqrt{2} \cdot \frac{1}{\sqrt{2}} + 0 \cdot \frac{1}{\sqrt{2}} & \sqrt{2} \cdot \frac{1}{\sqrt{2}} + 0 \cdot \left(-\frac{1}{\sqrt{2}}\right) \\ 0 \cdot \frac{1}{\sqrt{2}} + 0 \cdot \frac{1}{\sqrt{2}} & 0 \cdot \frac{1}{\sqrt{2}} + 0 \cdot \left(-\frac{1}{\sqrt{2}}\right) \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix} = A \end{aligned}$$



Summary Today (Write Your Own)

One sentence summary:

Detailed summary:



Summary Today (Write Your Own)

One sentence summary:

Detailed summary:



Summary Today (Write Your Own)

One sentence summary:

Detailed summary: